

Nadia Sultana

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About Me

I am interested in Explainable AI, Human-Computer Interaction, and biomedical machine learning. My research focuses on developing interpretable deep learning models for healthcare applications. I enjoy working on problems that make AI more understandable and useful for people, and I am looking to continue this research through a PhD.

Research Interests

Explainable AI (XAI), Human-Computer Interaction (HCI), Human-AI Interaction, Interpretable Machine Learning, Biomedical Signal & Image Analysis, Accessibility-Aware Intelligent Systems, Interactive Data Visualization

Education

M. Sc. in ICT (Thesis Defended)

Sep 2024 – Expected Jun 2026

Institute of Information Technology, Jahangirnagar University
CGPA: 3.71/4.00

B. Sc. in ICT, Institute of Information Technology, Jahangirnagar University
CGPA: 3.69/4.00

Feb 2019 – Sep 2024

Honors: Merit-Based Scholarship (for Admission Test, 1st Year, and 3rd Year academic results)

Research Grants & Fellowships

National Science & Technology (NST) Fellowship | Ministry of Science & Technology, Bangladesh

2025–26

Awarded for M.Sc. research on explainable deep learning for Parkinson's Disease diagnosis.

Grant Amount: 54,000 BDT

Journal Publications

- **Nadia Sultana**, Sidratul Afrida, Bushra Akter, Tasnia Jahan, Afrin Ahmed, Mohammad Abu Yousuf, and Md Zia Uddin. “**Explainable CNN-GRU Model with Self-Attention for Human Activity Recognition Using Wearable Sensors through Data Augmentation**,” in *IEEE Access*, vol. 14, pp. 48448-48477, 2026, **Q1 Journal** [Published] (Based on B.Sc. Thesis)
- Sanjida Akter, Md. Akmol Masud, Mst. Sanjida Islam Promi, **Nadia Sultana**, Maruf Ahmed, Md. Mahmudur Rahman, Mohammad Abu Yousuf, Saad Aloteibi, and Mohammad Ali Moni, “**DREAM: A novel explainable neural network for detecting sleep apnea using single-lead ECG signals**,” in *Biomedical Signal Processing and Control (Elsevier)*, vol. 114, 109291, 2026. **Q1 Journal** [Published]
- Md. Akmol Masud, Sanjida Akter, **Nadia Sultana**, Mohammad Shahidul Islam, Mohammed Abu Yousuf, Farzan M. Noori, and Md Zia Uddin “**MosQNet-SA: Explainable convolutional-attention network for mosquito classification with application as a RESTful API for dengue and malaria risk mapping**,” in *PLoS ONE*, vol. 21, pp. 1-30, 2026. **Q1 Journal** [Published]
- Md. Akmol Masud, Sanjida Akter, **Nadia Sultana**, Md. Biplob Hosen, Mehrin Anannya, Mohammad Abu Yousuf, AKM Azad, Salem Alyami, and Mohammad Ali Moni. **Pseudo-RGB data augmentation with deep-learning algorithms for improved Alzheimer's disease and other disabilities detection**. *PLoS ONE* [Under review]

Conference Papers

- Md. Akmol Masud, Sanjida Akter, **Nadia Sultana**, Mohammad Abu Yousuf, and Md Zia Uddin, “**Multi-layered password-based steganography: A novel approach for tiered information hiding**,” in *Proceedings of the 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, pp. 1–6, IEEE, Jan 2025 [Published]
- **Nadia Sultana**, Md. Tahmidur Rahman, and Fahima Tabassum. “**Incep-BiLSTM-Attn: An Explainable Framework for Parkinson's Diagnosis Using Voice Biomarkers**” in *The 2nd International Conference on Federated Learning and Intelligent Computing Systems (FLICS2026)*, June 2026 [Accepted] (Based on M.Sc. Thesis; to be presented at conference)

Professional Experience

Associate UX Engineer, Synesis IT PLC. – Dhaka, Bangladesh

Aug 2024 – Present

- Contributed to 15+ digital platforms, including 10+ national e-government initiatives across telecommunications, finance, agriculture, taxation, and citizen services.
- Led UX design, requirement analysis, wireframing, and frontend implementation for multi-module government systems serving large-scale public users.
- Built reusable Figma component libraries and scalable design systems to maintain consistency across complex workflows.
- Collaborated with cross-functional teams, senior government officials, and developers to align business requirements with accessible, user-centered digital experiences.
- **Key projects:** Bancassurance System (Jiban Bima Corporation), 333 Dashboard & Portal (a2i), NEIR Portal (BTRC), Land Contact Center (Ministry of Land), National Job Portal (BCC), E-Return System (NBR), and Plant Quarantine System(DAE).

Technical Skills

- **Programming Languages:** C, C++, Python, Java, C#, MATLAB, SQL
- **Frameworks & Libraries:** TensorFlow, Keras, Scikit-learn, Django, Spring Boot, React, Unity
- **Web & Application Development:** HTML, CSS, JavaScript, Frontend Development, Backend Development, Android Studio
- **Design & Prototyping:** Figma, Adobe Illustrator, WordPress

Training & Certification

Training of Trainers (ToT): B-TopSE Program Agile 2 (Agile Product Development)

Dec 2025 – Jan 2026

JICA, BCC, BASIS TCP

- Completed a Training of Trainers program on Agile product development, covering agile methodologies, team facilitation, and agile project management practices.

B-TopSE Program: Software Architecture Course

Sep 2025 - Oct 2025

JICA, BCC, BASIS TCP

- Completed intensive training on modern software architecture and system design methodologies, including UML structural modeling and the ICONIX process (Use Case → Domain → Robustness → Design), focusing on scalable system architecture.

Projects

Human Activity Recognition Using ML (UCI-HAR Dataset)

github

- Built a system to classify human activities from sensor data using Logistic Regression, SVM, Random Forest, and KNN.

Predicting Coronary Disease Using Logistic Regression (Framingham Dataset)

github

- Analyzed patient data to predict coronary disease risk by performing data preprocessing, feature selection, model training, and evaluation to identify key risk factors using logistic regression.

🔗 Explore more projects on GitHub

Academic Service & Leadership

Head of Department – Graphic Design, Youth School for Social Entrepreneurs

Oct 2021 – Jul 2022

- Led recruitment, supervised daily operations, and managed a team of **40+ members**, contributing to the team's recognition as the **Best Department**.

Vice President (IT), E-Business & Entrepreneurship Club, Jahangirnagar University

Aug 2022 – Sep 2024

- Oversaw IT operations, including the club's website and social media, while facilitating networking and improving programs to promote the club's mission.

References

Dr. Mohammad Abu Yousuf

Vice-Chancellor, University of Frontier Technology, Bangladesh

Professor, Institute of Information Technology, Jahangirnagar University, Bangladesh

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